This line declares that the first variable is named `red`; identifies the variable's data type as an integer, or `int`; and then tells the compiler that this variable should stay constant using the `const` modifier. Finally, this one line of code will also assign the numerical value `9` to the variable named `red`. In this way, the keywords `red`, `green`, `blue` are all variable names that we made up to identify the pin that each color of the RGB LED is connected to, while `time` is a variable to store the time period that each color stays on for, and `multiple` will be used to delay for a longer period on white just to give it a little emphasis.

Later in the sketch, we substitute the numbers that might be found inside the various statements with the variable names that we set up earlier so that every time that name is called, the compiler knows to look up the appropriate value. By using variables in our code, the following two lines will be functionally equivalent:

```
pinMode(red, OUTPUT);

pinMode(9, OUTPUT);
```

The rest of the code is just a slightly longer version of our last project sketch. In addition to red, green, and blue, you should now see yellow, cyan, magenta, and even white. Pretty nifty, huh? In addition to turning on and off each color of LED, we turn on two colors at the same time to make the intermediary colors, and all three colors to make white. Inside the `loop()` function, we begin with red, then add green to make yellow, turn off red to make green, add blue to make cyan, turn off green to make blue, turn on red to make magenta, and then turn on green to make white. At this point, we perform a compound operation to multiply the time delay with another variable called `multiple` to create a longer delay on the white color. We then reset our colors by turning off blue and green, leaving only red, perform another operation on the time delay to return it back to its original value, and then we start the loop all over again.

Now that we have seen variables in action, let's look at them in much greater depth to find out how we can put them to work for us.

What's a Variable?

The variables that we use in writing code are the very same variables that we learned about in school. Take the following equation for example:

$a^2 + b^2 = c^2$

I'm sure you've seen this before; it's the Pythagorean theorem and it's pretty useful for finding the length of one side of a right-angle triangle if we know any of the other two sides. The letters `a`, `b`, and `c` are the names for numerical data, in this case the letter `c` is used for the hypotenuse while `a` and `b` are the other two sides of the triangle. We can also call these letters variables. **Variables** are quite simply containers for information. In programming, variables have a **name** that tells us what data the variable might contain and they have a **type** that determines what kind of information the variable can hold, and finally they have a **value** whether we know what it is or not.

We can use variables to assign human-readable names for things like pin numbers or useful values. They can also be used to store numbers that constantly change while our code is actively running. This can be useful for setting up counters, performing calculations, or for passing new values to and from functions. To properly use a variable, we need to start with declaring the variable. After that, we should also talk about good variable names, the various types of variables, and a few other things related to working with variables.